

Blockchain-Assisted Mission Trust in Air–Surface–Underwater Unmanned Systems: A Cross-Layer Review and Object–Assurance–Lifecycle Framework

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Abstract

Air–surface–underwater missions combine unmanned aerial, surface, and underwater vehicles across heterogeneous links, intermittent connectivity, and administrative domains. A mission decision must combine evidence about physical observations, digital records, shared state, and fitness for use. This narrative review synthesizes literature on cross-domain cooperation, unmanned-system security, trust and reputation, provenance, remote attestation, zero trust, and permissioned blockchains. The synthesis introduces the Object–Assurance–Lifecycle (OAL) framework, relating four trust objects (entity, platform/device, link/path, and data product) to four assurance layers (observation validity, commitment and provenance integrity, ledger and governance consistency, and mission-use suitability) across six mission stages. A typed dependency graph and context–temporal–graph reasoning loop separate evidence appraisal from reliance policy and identify where lineage, common causes, and missing evidence affect decisions. A worked example illustrates the resulting acceptance and restriction rules. The synthesis identifies conditional ledger roles in membership, commitments, version coordination, audit, and post-partition reconciliation, with physical validity, endpoint protection, local safety, and governance supported by complementary controls. A claim-specific evidence-maturity profile, research propositions, and a proposed Cross-Domain Unmanned Trust Profile define an evaluation agenda. This conceptual framework provides a basis for empirical evaluation of comparative performance and integrated mission effectiveness.

Keywords: mission trust; unmanned systems; UAV–USV–UUV cooperation; blockchain; data provenance; remote attestation; zero trust; maritime autonomy

1. Introduction

Unmanned maritime operations are becoming systems of systems. A UAV can rapidly survey a wide area and maintain a radio link; a USV can loiter, carry compute and communications equipment, and bridge air and underwater segments; a UUV can collect close-range observations beyond the reach of radio and satellite services. Cross-domain control studies increasingly coordinate such platforms for search, tracking, formation, navigation, and surveillance missions [1–7]. These complementary capabilities link domains with substantially different communication physics, failure modes, owners, and security assumptions. Broad reviews of UAV networks and applications emphasize mobility, limited energy, intermittent links, and safety constraints [8–10]; reviews of unmanned marine and underwater vehicles add acoustic delay, difficult localization, harsh deployment conditions, and

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constrained recovery [11–13]. A mission decision may therefore depend on evidence that has crossed several media, gateways, processing stages, and administrative boundaries.

This creates a *mission-trust gap*. Conventional cybersecurity questions—who authenticated, which packet arrived, whether a file hash matches, or whether replicas agree—each characterize a specific assurance property; mission-use suitability requires these properties to be composed for a particular decision at a particular time. A correctly signed sonar report can be physically implausible; a valid observation can be replayed after it becomes stale; a correctly ordered ledger entry can originate from a captured endpoint; and a well-provenanced product can still be unsuitable for a high-consequence task because its uncertainty is too large. Research on UAV and swarm security catalogs spoofing, jamming, routing attacks, capture, malware, data manipulation, and privacy threats [14–17]. Underwater-network security reviews describe related attacks under low bandwidth, long propagation delay, and expensive retransmission [18–21]. These literatures provide the mechanisms from which an end-to-end chain from physical observation to mission use can be constructed.

Blockchain is frequently proposed as an answer to distributed trust. UAV-oriented reviews report uses in identity, authentication, coordination, access control, data sharing, and audit [22–26]. Representative proposals use ledgers for swarm identity, lightweight authentication, trusted message exchange, and distributed data acquisition [27–31]. Maritime reviews similarly associate blockchain with documentation, information sharing, authentication, and inter-organizational coordination [32–34]. A shared, append-oriented record allows participating organizations to exercise joint control over its history. Precise use of the word *trust* distinguishes replicated-state consistency from sensor truth, endpoint integrity, privacy, and timeliness. General blockchain security and Internet-of-Things (IoT) surveys document the associated scalability, resource, governance, privacy, and interoperability conditions [35–38]. The resulting analytical task is to map ledger functions to the claims they support, the complementary controls they require, and the deployment conditions under which shared state adds value.

Adjacent research offers parts of the answer. Zero trust bases least-privilege access on explicit, continuously evaluated evidence about subjects, devices, and resources [39,40]. Remote attestation separates the production of device evidence, its appraisal, and the relying party's decision [41–43]. Provenance models describe how entities, activities, and agents are related and can support accountability across transformations [44–47]. Trust-management research supplies reputation, probabilistic, subjective-logic, fuzzy, and learning-based methods [48–53]. These research strands describe their mechanisms in distinct architectural vocabularies. A cross-domain mission requires them to be aligned around the objects being judged, the assurance claim being made, and the stage at which evidence is needed.

This review advances one central argument: *mission trust rests on an assurance chain whose distinct claims each require appropriate evidence, with blockchain strengthening selected transitions within that chain*. The proposed *Object–Assurance–Lifecycle* (OAL) framework structures this argument for mission analysis and design. It relates four trust objects—entity, platform/device, link/path, and data product—to four assurance layers—observation validity, commitment and provenance integrity, ledger and governance consistency, and mission-use suitability—over six lifecycle stages from admission to recovery. A cross-cutting assurance and governance plane coordinates identities, attestation results, provenance commitments, trust state, policy and model versions, ledger state, and consortium rules across the four mission objects.

The synthesis has three connected outputs. First, OAL provides a common structure for comparing mechanisms by object, assurance claim, and lifecycle stage. Second, a typed

dependency graph and context–temporal–graph (CTG) loop organize evidence appraisal, uncertainty, affected descendants, policy action, and audit. Their threat grammar records attacker, capability, object, propagation path, mission impact, and control, while distinguishing causal hypotheses from insufficient evidence. Third, an evidence-maturity profile and the proposed Cross-Domain Unmanned Trust Profile (CDUTP) translate this structure into claim-specific evaluation requirements and an interoperability agenda. These analytical proposals build on established zero-trust, attestation, provenance, communications, and ledger mechanisms and define questions for evaluating usefulness and implementation cost.

The analysis focuses on civil applications and publicly documented defense-support settings, where coalition ownership, contested communication, and mission criticality make governance especially consequential. This public-evidence scope supports reproducible analysis of assurance architecture, evidence semantics, failure modes, and evaluation methods. The contribution is a conceptual integration, a set of technology-selection boundaries, and an evaluation agenda for future empirical and formal studies.

The remainder of the paper describes the review scope and analytical lens (Section 2), establishes the operational and conceptual foundations, compares complementary research landscapes, develops OAL and CTG reasoning, examines blockchain patterns and boundaries, organizes evaluation evidence by maturity, and closes with research propositions, scope and applicability, and conclusions.

2. Review Scope and Analytical Lens

This narrative review draws on 141 references: 130 scholarly papers and book chapters, one book, and 10 standards, technical reports, or other official technical references. The literature is in English and mainly covers publications from 2016–2026, supplemented by earlier foundational work on reputation, distributed-system limits, provenance, and consensus. The scholarly literature includes original studies, reviews, and papers describing datasets, benchmarks, and simulation platforms.

The references cover six related themes: UAV blockchain and swarm security; maritime and underwater communication and security; heterogeneous UAV–USV–UUV cooperation; IoT trust and reputation; provenance, attestation, and zero trust; and permissioned-ledger architecture and governance. Unmanned-system studies establish the operational constraints of air, surface, and underwater missions. General IoT, cybersecurity, and distributed-systems research provides mechanisms for authentication, evidence appraisal, data sharing, and coordination. Standards and official specifications supply architectural definitions and interoperability requirements, while datasets and simulation platforms provide resources for evaluation.

The analysis compares mechanisms by the object being assessed, the assurance claim they support, and the mission stage at which they operate. These three dimensions form the Object–Assurance–Lifecycle (OAL) framework. Each comparison considers the supporting evidence and the network, resource, threat, and governance assumptions that condition its use. Transfer to a maritime mission depends on whether those assumptions hold in the target environment. For example, a consensus mechanism evaluated on shore servers requires separate assessment under the energy limits and intermittent acoustic connectivity of a UUV. This perspective connects the literature to the dependency analysis, CTG reasoning loop, and evaluation agenda developed in later sections.

Table 1 summarizes the review scope. UAVs, USVs, and underwater vehicles form the operational focus, with satellite, shore, cloud, and edge systems providing supporting infrastructure. The lifecycle extends from admission and sensing through mission decisions

to audit, revocation, and recovery, encompassing both operational trust and accountability across participating organizations.

Table 1. Review scope and analytical focus.

Dimension	Analytical focus	Complementary context
Physical domains	UAV, USV, UUV / AUV / ROV; buoys and cross-media gateways	Ground-vehicle analogies and platform structural-certification requirements
Infrastructure	Satellite, high-altitude, shore, cloud, edge, and consortium services	Partition-aware operation and explicit infrastructure trust assumptions
Mission settings	Observation, inspection, search and rescue, environmental monitoring, logistics, and publicly documented defense-support contexts	Publicly reportable assurance semantics, failure modes, and evaluation methods
Trust mechanisms	Identity, authorization, attestation, provenance, dynamic inference, policy, ledger, audit, and governance	Resource requirements, evidence freshness, uncertainty, and conditions for cross-domain use
Evidence	Original studies, reviews, a scholarly book, standards, and official technical references	Datasets, benchmarks, simulation platforms, and foundational work from related fields
Review output	Conceptual framework, taxonomy, design boundaries, evidence profile, and research propositions	Empirical comparison, formal verification, and deployment certification as evaluation stages

3. Operational Context and Trust Foundations

3.1. A Heterogeneous Mission System of Systems

The core system contains UAVs, USVs, and UUVs that contribute different sensing, mobility, computation, and communications functions. UAVs offer rapid repositioning, line-of-sight radio relays, and electro-optical sensing, but their payload, endurance, and link stability are constrained. USVs can carry larger energy stores and antennas, remain on station, and bridge acoustic and radio networks. UUVs provide proximity to underwater targets and environments but frequently depend on low-rate acoustic channels, short-range optical links, or delayed surfacing for high-volume transfer. Surveys of UAV-aided maritime communication and 6G maritime connectivity illustrate the relay opportunities [54,55]; underwater communication reviews document the trade-offs among acoustic, optical, and other media [56–59]. The resulting system requires a dynamic network model in which topology, latency, bandwidth, clock quality, and reachability can change during a single mission.

Communication requirements in civil UAV missions vary with network role, data volume, range, and quality-of-service needs [8]. In maritime search, heterogeneous co-operative systems remain at an early stage of application despite increasingly mature UAV, USV, and UUV capabilities [60]. Mission trust is therefore conditioned on current role and mission phase, with evidence re-evaluated whenever the application domain or consequence changes.

Cross-domain cooperation also changes fault containment. A USV may be the navigation reference, data cache, radio gateway, and permissioned-ledger peer for several UUVs. Compromise or benign failure of that high-betweenness platform therefore affects link evidence, time, provenance continuity, and access to the shared record. Conversely, platform

diversity can improve resilience when evidence from independent media and locations can corroborate an event. Cooperation research demonstrates the control and planning value of heterogeneous formations and multi-swarms [5,6,61,62]. Mission assurance extends these results by making the trustworthiness of shared observations, commands, and dependencies explicit.

Administrative heterogeneity is equally important. A civil emergency may combine government responders, commercial communications providers, research vehicles, and contracted data services. A coalition mission may include platforms that remain under distinct authorities. Cross-organizational authority therefore combines local authentication with policy equivalence, delegated roles, and accountable governance. The system boundary includes organizations, credentials, data-use restrictions, software and model versions, and the process that admits, suspends, or removes participants. Under these conditions, a shared ledger can provide a consistent audit state whose history is jointly governed by the participating parties.

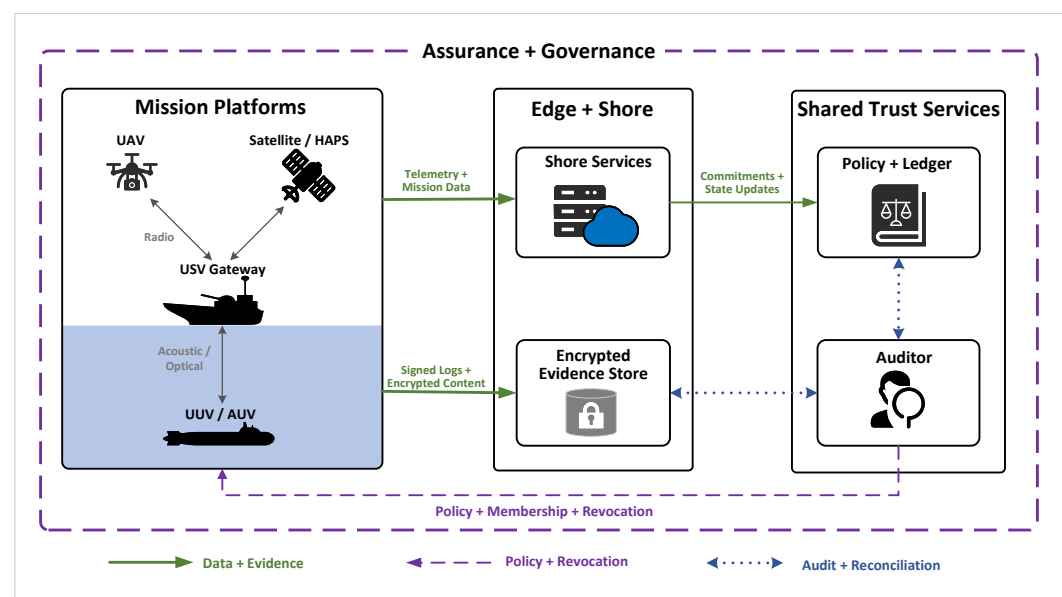


Figure 1. Proposed cross-domain reference architecture. Solid teal arrows show data and evidence moving toward shore and shared services; dashed violet arrows show policy, membership, and revocation returning to mission platforms; dotted slate links show audit and reconciliation. Radio and acoustic/optical media are labeled on neutral topology lines. Constrained platforms retain signed local logs, while capable gateways and shore services forward commitments when connectivity permits. The dashed enclosure identifies the cross-cutting assurance and governance plane.

Figure 1 presents an adaptable logical deployment. Raw imagery, sonar, telemetry, and detailed attestations remain encrypted off chain. A permissioned ledger may retain content identifiers, commitments, coarse-grained state transitions, membership changes, policy and model versions, revocations, and audit decisions. During disconnection, an edge node keeps a signed, hash-linked sequence and later submits a batch commitment. Reconciliation can expose gaps and conflicts when a retained checkpoint, counter, or independent witness makes them observable. An event omitted before commitment, or an unwitnessed deleted suffix, may remain undetectable. Evidence retention and retrieval therefore complement record integrity; local sensing and control mechanisms retain responsibility for physical interpretation and real-time deadlines.

3.2. Mission Trust, Trustworthiness, and a Trusted Data Environment

Mission trust is defined here as a context-specific relying decision about whether an object may contribute to a stated mission purpose, given available evidence and the

consequences of error. Trust is therefore relational and prospective: a consumer relies on an object for an action. *Trustworthiness* denotes properties that justify such reliance, including identity authenticity, platform integrity, link availability, provenance, freshness, data quality, and accountable governance. Both vary with time and context. The same UAV may be suitable for low-risk environmental imagery while being excluded from time-critical navigation support because its attestation is stale or its location is uncertain.

For an object o , time t , context c , and mission use m , evidence assessment and the relying decision have distinct interfaces:

$$\begin{aligned} T_o(t, c, m) &= \mathcal{A}(o, \mathcal{E}_{\leq t}, c, m; \theta), \\ D_{r,o}(t, c, m) &= \pi_r(T_o(t, c, m), \mathcal{C}_t, m). \end{aligned} \quad (1)$$

Here $\mathcal{E}_{\leq t}$ is the evidence received by decision time, including its observation times and lineage; $\mathcal{A}$ is an appraisal procedure with assumptions or parameters θ ; and T_o retains claim-specific outcomes and uncertainty. The relying party r applies policy π_r under constraints $\mathcal{C}_t$, including authority, deadlines, consequences, and available alternatives. $D_{r,o}$ may permit, restrict, defer, or reject a use. Two relying parties can consequently choose different actions from the same evidence.

One implementation may estimate $\Pr(H_{o,m} \mid \mathcal{E}_{\leq t}, c; \theta)$ for an explicitly defined hypothesis $H_{o,m}$ about required properties. That interpretation requires a specified probabilistic model and calibration evidence. Subjective opinions, intervals, ordinal states, and reputation scores provide other representations [51–53]. A probabilistic interpretation of a normalized score requires an explicit event model and calibration evidence. Equation (1) specifies interfaces for appraisal and decision, leaving their implementation and validation to a concrete system.

A *trusted data environment* is the socio-technical setting that produces, transports, transforms, shares, and governs evidence and mission data. It includes principals, devices, paths, data products, policies, models, services, and accountability processes. Mission trust is the decision relation exercised within that environment. A tamper-evident catalog strengthens the environment, while mission-use appraisal determines which cataloged item is fit for a particular decision. Zero trust contributes the access-control architecture: NIST's formulation uses policy engines, identity and device evidence, enforcement points, and protected resources [39]. Intermittently connected missions add safe degradation and visible evidence staleness to those functions.

3.3. Four Complementary Assurance Claims

The assurance chain in Figure 2 distinguishes four questions. Observation validity asks whether a measurement plausibly represents the physical phenomenon, considering calibration, geometry, environment, sensor health, and corroboration. Commitment and provenance integrity asks which producer, time, place, and transformation claims are bound to the item, and whether the recorded bytes and lineage were altered. Ledger and governance consistency asks whether authorized record keepers agree on membership, order, versions, and state transitions under stated fault assumptions. Mission-use suitability asks whether the item is timely, relevant, sufficiently certain, policy-compliant, and proportionate to the consequences of the contemplated action. These complementary assurance categories allow implementation-specific evaluation sequences; ledger-specific claims apply to architectures that use a ledger.

These claims have distinct failure modes. GNSS spoofing can make an otherwise healthy UAV report the wrong location; machine-learning and multi-sensor approaches may detect inconsistency, with validity conditioned on data and context [63–67]. A stolen signing key can produce syntactically valid provenance. A permissioned consensus group

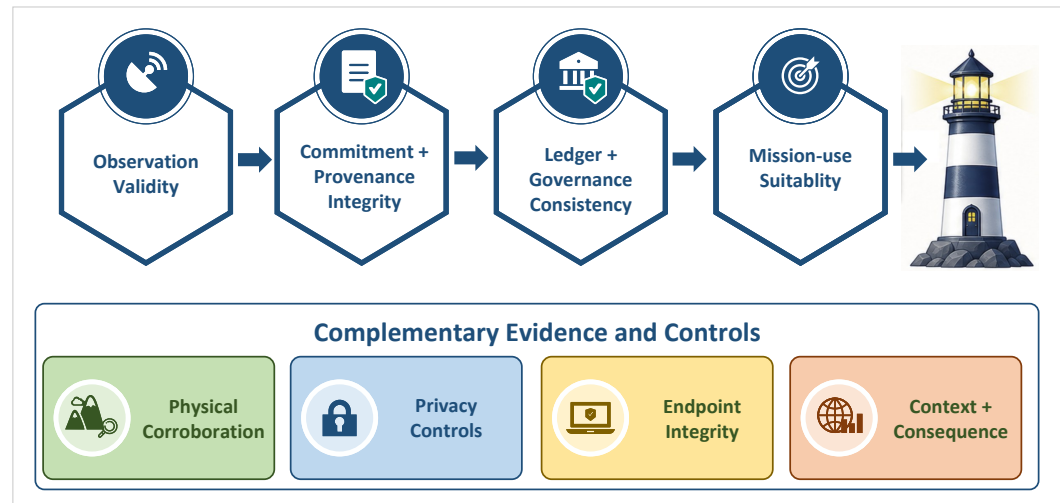


Figure 2. Four-layer assurance chain. Each applicable layer contributes a distinct claim to the relying decision. Arrows organize the argument; logical relationships require explicit inference rules, and execution order depends on the implementation.

can faithfully order false submissions. A delayed but accurate observation can be unfit for collision avoidance. The chain therefore keeps four concepts distinct: signatures and physical truth; hashes and privacy; consensus and endpoint security; and historical reputation and current mission fitness.

Remote attestation and device baselines fit between the first two layers. The RATS architecture distinguishes evidence, appraisal policy, attestation results, and relying parties [41]; the NIST IoT baseline emphasizes device identification, configuration, data protection, logical access, software update, and cybersecurity-state awareness [68]. These mechanisms justify scoped claims about measured software and device state through roots of trust, reference values, verifier freshness, and explicit measurement coverage. External observation validity is established through complementary physical and sensor evidence. Provenance records derivation: W3C PROV relates entities, activities, and agents, enabling consumers to ask how a product was generated [44,45]. OAL brings these distinct claims into one mission-oriented vocabulary.

4. Complementary Research Landscapes

The relevant literature addresses communication performance, cooperative control, adversarial behavior, trust estimation, provenance, and shared-state management. Table 2 maps representative contributions to the mission-assurance questions they inform. These relationships identify mechanisms that can be combined and the assumptions governing their use in air–surface–underwater missions.

4.1. Cross-Domain Cooperation and Assurance Semantics

Cooperation studies clarify what platforms can do together. They address formation control, cooperative path planning, and target search across heterogeneous vehicles [2–5,7,62]. In these control studies, *consensus* denotes convergence or coordination of agent states; replicated-ledger consensus concerns agreement on a shared record state under a specified fault model. The corresponding guarantees apply to different objects and assumptions. OAL proposes to connect a coordination output to security provenance, delegated authority, and cross-organizational audit by treating the output as a data product with traceable inputs, transformation, policy version, and intended mission use.

Maritime search and unmanned-vehicle reviews connect mission capability to sensing, localization, communication, and planning [11,13,60]. These dependencies motivate claim-

Table 2. Representative research strands and their role in the OAL synthesis. The integration questions connect the cited research focus to cross-domain assurance requirements.

Research strand	Source focus	Question for cross-domain integration	Representative sources
UAV networks and security	Mission-dependent communication requirements; platform, software, and network threats	Which endpoint and path assumptions accompany an observation or command?	[8,14–16]
UAV blockchain	Ledger applications, architectures, opportunities, and resource constraints	Which admission, data-sharing, or audit claim requires shared administration?	[22–25]
Marine and cross-domain cooperation	Vehicle capabilities, cooperative search, and communication support	How are sensing, planning, and gateway dependencies carried into a relying decision?	[11,12,54,60]
Underwater communication and security	Channel constraints, reliable collection, threats, and countermeasures	How should delayed or missing evidence affect reliance under both attack and degradation?	[18,21,59,69]
IoT trust and reputation	Evidence aggregation, recommendations, context, and reputation management	What property does an estimate measure, and when does that estimate support mission use?	[48–50,70]
Provenance, attestation, and zero trust	Data lineage, trust anchors, device appraisal, and resource-access policy	How are origin, measured state, and authorization bound to the same product and decision time?	[40,46,47,71]
General IoT blockchain	Integration architectures and computation, storage, communication, and security trade-offs	Which records benefit from replicated governance, and which evidence remains outside the ledger?	[36,37,72,73]

specific evidence across platform, path, and data-product states. As illustrative failure cases, a UAV may be healthy while its acoustic path is jammed; a USV gateway may authenticate yet relay a stale model; and UAV imagery may retain authentic pixels while its georegistration is wrong. Each relying decision therefore requires the relevant combination of claims, with the consequences of a failed claim defined for the intended use.

4.2. From Security Catalogs to Cross-Object Propagation

UAV, swarm, and underwater-security reviews organize attacks by target and mechanism. Across these domains, reported categories include physical capture, identity and authentication attacks, GNSS spoofing where satellite positioning is used, jamming, denial of service, malicious routing, eavesdropping, malware, false-data injection, and privacy leakage [14,15,17–19]. Hierarchical detection and response provides an example of distributing monitoring and countermeasures in a UAV network [74]. For OAL, dependency analysis asks how an initial compromise affects subsequent claims. If a captured gateway relays traffic, transforms observations, and holds batch-endorsement keys, a valid ledger commitment may preserve its manipulated output. Effective controls then span the endpoint, transformation, key custody, and ledger interface.

Environmental degradation creates an additional classification problem. Acoustic multipath, Doppler, noise, and changing topology can produce loss, latency, or disagreement; optical links introduce different attenuation and alignment constraints [56,57,75,76]. Such effects can resemble attack. A trust system that equates anomalies with malice may isolate useful nodes during severe conditions, while an adversary may exploit expected environmental variation. A cross-object dependency model should therefore consider both malicious manipulation and benign degradation, including their coexistence, and retain an unknown status while the causal explanation remains unresolved. Abstention or restricted use can follow directly from the assessed uncertainty and mission consequence; attribution of malicious intent requires its own evidence.

4.3. Trust Computation with Mission Semantics

IoT trust and reputation literature provides taxonomies of direct observation, recommendation, context, centralized and distributed management, and attacks on reputation [48–50]; Internet-of-Vehicles work supplies related models under a different mobility and communication setting [77]. Underwater studies have explored decision-tree and hidden-Markov approaches [78,79], and design guidance recognizes underwater-specific evidence constraints [80]. These methods produce classifications or estimates under their model assumptions. Interpreting an output as a calibrated probability of safe mission use requires

additional validation. OAL requires each estimate—for example, routing reliability, current software integrity, sensor accuracy, or willingness to cooperate—to name the object, property, context, evidence lineage, uncertainty expression, and permitted use.

Blockchain-based trust-management studies examine shared reputation and tamper-evident updates for IoT and distributed services [70,81,82], with related designs reviewed in cloud computing [83]. A ledger can make accepted updates auditable under its membership, key-management, and consensus assumptions. It can also preserve misleading recommendations. Collusion, correlated observations, model drift, and incorrect identity binding require evidence appraisal and model validation beyond record integrity. Storage and governance choices therefore follow the definition of the claim and its evidence semantics.

Fog and edge trust approaches place appraisal near constrained devices, while trust-recommendation reviews examine context, recommendation quality, and attack resistance [84,85]. In a cross-domain mission, a USV edge service may observe several UUVs while remaining their common gateway. Local appraisal can avoid remote communication round trips, but recommendations from that service share dependencies on its platform, clock, path, and administrative owner. Recording those dependencies makes common causes explicit so that aggregation can avoid treating several reports as independent corroboration solely because they concern different vehicles.

4.4. Provenance, Attestation, and Authorization

Provenance research provides a vocabulary for origins, transformations, and responsibility [46,47]. W3C PROV represents entities, activities, and agents, with relations for generation, use, derivation, and attribution [44]. Its terminology needs an explicit mapping: an OAL entity is an acting principal, usually represented as a PROV agent, whereas an OAL data product corresponds to a PROV entity. A provenance description records asserted lineage; reliance on that description additionally requires evidence of trustworthy capture, cryptographic binding, and availability of the referenced records.

Remote attestation addresses a different claim: a verifier appraises evidence about an endpoint, and a relying party uses the resulting assessment. RATS makes the appraisal policy, trust relationships, and freshness conditions explicit [41]; trust-anchor surveys describe the supporting technologies [71]. Zero trust supplies principles for resource-specific authentication and authorization using subject, device, and contextual information [39,40]. OAL connects these mechanisms at decision time: provenance identifies the input and transformation, attestation contributes measured endpoint-state evidence, and authorization identifies permitted reliance. Sensor validity and fitness for a particular physical action still require their own evidence. During disconnection, the integration must specify how long cached assessments and delegated permissions remain usable.

4.5. Ledger Mechanisms and Physical-Evidence Boundaries

Blockchain surveys and technical guidance explain architectures, consensus mechanisms, smart contracts, scalability, and security [86–90]. IoT integrations confront limited computation, storage, communication, energy, privacy, and interoperability [36,37,91]. UAV proposals explore authentication and trusted swarm or network architectures [31,92–94]; maritime transportation proposals address authentication and aggregate signcryption [95,96]. These precedents inform choices for membership, key custody, endorsement, validation, and on-chain/off-chain binding. Their relevance to cross-domain missions depends on the tested deployment, connectivity, and threat assumptions. Establishing operational integration across UAVs, USVs, and UUVs requires evaluation of the combined deployment.

Cross-sector classifications discuss scalability, integration, and governance problems [97]. Technical guidance also identifies the boundary between recording sensor input and establishing whether it reflects a physical event [86]. UAV-specific taxonomies and 6G perspectives motivate evaluating ledger placement against resource, mobility, and network-architecture constraints [98–100]. For the present synthesis, these observations motivate evaluating capable gateways and edge nodes as ledger hosts, together with the new availability and administrative dependencies this placement creates. OAL provides a way to specify the mission-assurance claim that would justify that cost and to compare it with an authenticated non-ledger alternative.

The synthesis opportunity is a common analytical model connecting a physical observation, the responsible principals and devices, the paths and transformations, the shared governance record, and the eventual mission decision. The next section develops that model.

5. The OAL Framework for Mission Trust

The *Object–Assurance–Lifecycle* (OAL) framework provides a three-dimensional structure for classification and design. It can be used to locate a published mechanism, specify an assurance requirement, or identify an unprotected transition. The object dimension identifies *what is judged*; the assurance dimension identifies *which claim is sought*; and the lifecycle dimension identifies *when the claim is needed*. These axes are conceptually distinct; evidence and failures across their cells can be dependent. A mechanism may occupy several cells, and some object–claim combinations are inapplicable to a particular mission. The framework organizes an assurance case by locating applicable claims; completeness and effectiveness require evidence about the relevant dependencies and outcomes. Figure 3 presents the framework.

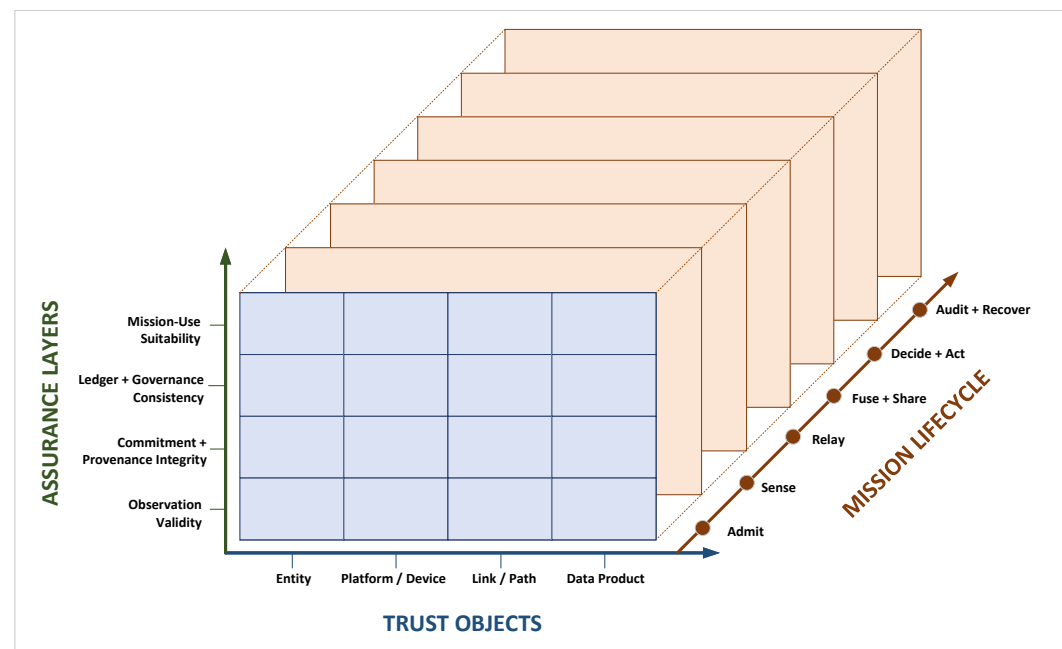


Figure 3. OAL framework for mission trust. Trust objects run along the horizontal front-face axis, assurance layers along the vertical axis, and lifecycle stages along the oblique depth axis. Cells distinguish claims and their timing; the cross-cutting assurance and governance plane supplies shared services and dependencies.

5.1. Dimension I: Four Trust Objects

An *entity* is a principal capable of holding authority or responsibility: an operator, organization, mission service, software agent, or autonomous decision-making principal. A *platform/device* is an executing physical or virtual endpoint: vehicle, sensor, gateway, compute module, secure element, or hosted execution instance. A stored software image or model artifact is a data product; its deployed state is assessed on the hosting endpoint. This distinction matters because an authorized entity can control a compromised device, while a device with valid attestation can execute a legitimate but unsuitable policy. An autonomous vehicle may occupy both categories under different claims: as a delegated principal for action and as an endpoint whose hardware and software state are measured.

A *link/path* is a time-varying end-to-end route composed of its physical media, intermediate nodes, security associations, and quality state. A data product may traverse a UUV acoustic hop, a USV gateway, a satellite backhaul, and a cloud broker; confidentiality, freshness, availability, and routing integrity depend on this composed path. A *data product* is an identified version of a raw observation, feature set, track, map, fused estimate, model output, command, alert, provenance record, or audit artifact. OAL treats a version as logically immutable: a transformation or correction creates a new identifier. This convention identifies versions; protection against storage alteration requires content verification and retained commitments.

Table 3. Four OAL trust objects, evidence, failure examples, and mission questions.

Object	Representative evidence	Illustrative failure or attack	Mission question
Entity	Credential chain, organizational role, delegation, authorization, prior accountable actions	Credential theft, insider misuse, collusion, obsolete authority	Who requests, endorses, or consumes the resource, and under which authority?
Platform/device	Secure boot measurement, firmware/software version, sensor self-test, configuration, energy and health state	Capture, malware, unsafe configuration, stale reference value, sensor drift	Which endpoint executes, and is its measured state adequate for this use?
Link/path	Authentication state, route, delay, loss, signal indicators, neighbor changes, medium and gateway state	Jamming, wormhole, selective forwarding, eavesdropping, benign fading	Through which changing path did evidence pass, and what security and quality properties held?
Data product	Signature, timestamp, location, provenance graph, quality meta-data, version, calibration and cross-source residuals	False injection, replay, poisoning, misregistration, unauthorized transformation	Where did this version come from, was it altered, and is it timely and suitable?

Policies, models, smart contracts, and ledger state map onto the four operational object categories. They are versioned artifacts hosted on devices or represented as data products, while the principals who authorize them remain entities. This stable object set accommodates new mechanisms, and the assurance plane makes their artifacts first-class subjects of versioning, access control, threat analysis, and audit.

5.2. Dimension II: Four Assurance Layers

The first layer, *observation validity*, asks whether evidence about the external world is credible. Relevant controls include calibration records, sensor diagnostics, geometry, physical bounds, environmental models, independent modalities, and temporal continuity. Multi-view and multi-sensor fusion can expose disagreement and express uncertainty [101–103]; it can also amplify correlated error or poisoned inputs. Observation assurance therefore documents independence assumptions and combines physical and domain evidence with cryptographic content commitments.

The second layer, *commitment and provenance integrity*, binds an item to recorded assertions about its producer, time, place, transformation, and version. Digital signatures, authenticated channels, secure timestamps, content identifiers, and W3C PROV relations support this layer [44–47,104]. A valid signature authenticates a statement under its key assumptions; the accuracy of the stated acquisition time or location requires separate evidence. Provenance covers both raw acquisition and model, human, or automated decisions. Decision provenance connects inputs, policy versions, intermediate transformations, and outputs, making later review possible when those records remain retrievable. Its assurance conditions include key custody, clock assumptions, instrumentation completeness, and trustworthy capture of events.

The third layer, *ledger and governance consistency*, concerns shared state and the rules by which administrative parties maintain it. It covers member admission, endorsement, record ordering, validation, version coordination, revocation, dispute handling, and reconciliation. Hyperledger Fabric illustrates an execute–order–validate architecture with identity-aware channels and endorsement policies [105,106]; other shared-state services can satisfy the applicable functions. A ledger can establish agreement on validated records under a declared fault model. The legitimacy of governance rules, resolution of organizational disputes, and semantic validity of submissions require additional evidence and procedures. For missions using another shared-state architecture, the corresponding database or audit controls define the applicable assurance claims; ledger-specific claims are marked inapplicable.

The fourth layer, *mission-use suitability*, applies a reliance policy to assurance evidence. Freshness, relevance, uncertainty, risk class, legal and organizational policy, available alternatives, and consequence severity are considered. A consumer may accept a low-resolution image for broad situational awareness but require corroborated, current, and accurately georegistered evidence for autonomous maneuver. An implementation assesses the required properties and separately applies the policy that permits, restricts, or defers an action. Authority to use the product requires the applicable authorization.

5.3. Dimension III: Six Lifecycle Stages

The lifecycle dimension prevents security from ending at admission. Table 4 shows how objects and assurance questions evolve from credentialing to recovery. Evidence can age, dependencies can change, and an initially valid decision can require reassessment.

Lifecycle continuity is especially difficult during partitions. During disconnection from a central policy engine, a UUV needs a local authorization envelope: named resources, permitted actions, risk limit, evidence freshness, expiry, and a context-appropriate fallback. Expiry enforcement requires bounded clock uncertainty and rollback-resistant local state, or a policy that reduces authority when those assumptions fail. Local actions are signed and sequenced; reconnecting nodes later disclose commitments and selected evidence. The design distinguishes *continued operation under delegated authority* from *fresh global authorization*; retained evidence supports later audit, while local enforcement constrains actions during disconnection.

5.4. The Assurance and Governance Plane

The cross-cutting plane contains identity and authorization, remote attestation, provenance capture, trust inference, policy/model/contract version management, ledger state, and consortium governance. Each component is itself attackable. Identity authorities can be compromised; reference values can be wrong; provenance instrumentation can omit transformations; a trust model can drift or be poisoned; a smart contract can encode an unsafe rule; validators can collude; and governance can deadlock. The plane must therefore be observed through the same OAL lens. For example, a model is a data product with

Table 4. Mission lifecycle and representative OAL assurance tasks.

Stage	Primary questions	Evidence and controls	Ledger-relevant state
Admission and authentication	Is the entity authorized, and is the joining endpoint in an acceptable measured state?	Credentials, delegation, attestation, configuration baseline, least privilege	Membership, role, attestation-result commitment, policy version
Sensing and collection	Does an observation plausibly reflect its stated time, place, modality, and calibration?	Sensor self-test, secure time/location, quality metadata, corroboration	Acquisition commitment and data identifier
Transmission and relay	Which path carried the item, and did quality or security conditions change?	Mutual authentication, route and link telemetry, delay/loss context, custody transfer	Signed relay events or batch roots
Processing, fusion, and sharing	Which inputs, code, model, parameters, and policies produced this version?	Provenance graph, reproducible transformation identifier, access and usage control	Version, endorsement, access decision, model/policy commitment
Decision and action	Is the product fit for the specific mission consequence now?	Uncertainty, freshness, risk policy, alternatives, human review or fail-safe	Decision rationale commitment and responsible principal
Audit, revocation, and recovery	What failed, who was affected, what must be invalidated, and how is operation restored?	Incident evidence, credential and model revocation, rollback, re-attestation, after-action review	Revocation state, dispute outcome, reconciled log, recovery version

provenance and validation evidence, a deployed instance is hosted on a device, its approval is an entity action, and its use is constrained by lifecycle stage and mission context.

OAL serves as both a diagnostic and a specification template. A blockchain-throughput study maps to ledger-performance cells and prompts complementary observation, endpoint, and mission-use requirements. A firmware-attestation protocol maps to a device claim at admission and prompts a freshness rule for disconnected decision time. For each applicable cell, a specification records the object/version, required property, evidence source and coverage, validity condition, acceptance rule, responsible party, and response to missing evidence. Reviewing identified high-consequence dependency paths then exposes absent or stale evidence. The resulting assurance case has the scope of its modeled paths; broader coverage requires identification and appraisal of additional dependencies.

6. Evidence Dependencies, Threat Propagation, and Dynamic Reasoning

6.1. A Typed Object-Dependency Graph

OAL classifies claims; a dependency graph records relations relevant to their appraisal. Let $G_t = (V_t, E_t)$ be a time-indexed directed multigraph whose vertices denote the four object types, with version or state-interval identifiers where applicable. Relations have declared source and target types: an entity *controls* a platform; a platform *hosts* another device or a model artifact; a platform *produces* an observation product; a path *transmits* a product; an entity *endorses* or *consumes* a product; and a ledger-record product *commits to* another product. A derived product *depends on* input products through an identified transformation. Each transformation identifier links all inputs, outputs, the executing platform, responsible entity, code/model version, and event-time interval. A W3C PROV serialization can represent that transformation as an activity node [44]; activities describe relations among the four OAL object types. An OAL entity corresponds to a responsible principal, usually a PROV agent, while a data product corresponds to a PROV entity. Time

and version qualifiers are essential. A decision based on model version 4.2 before a rollback requires that version and activity interval to qualify the relation “UAV-7 hosts model M.”

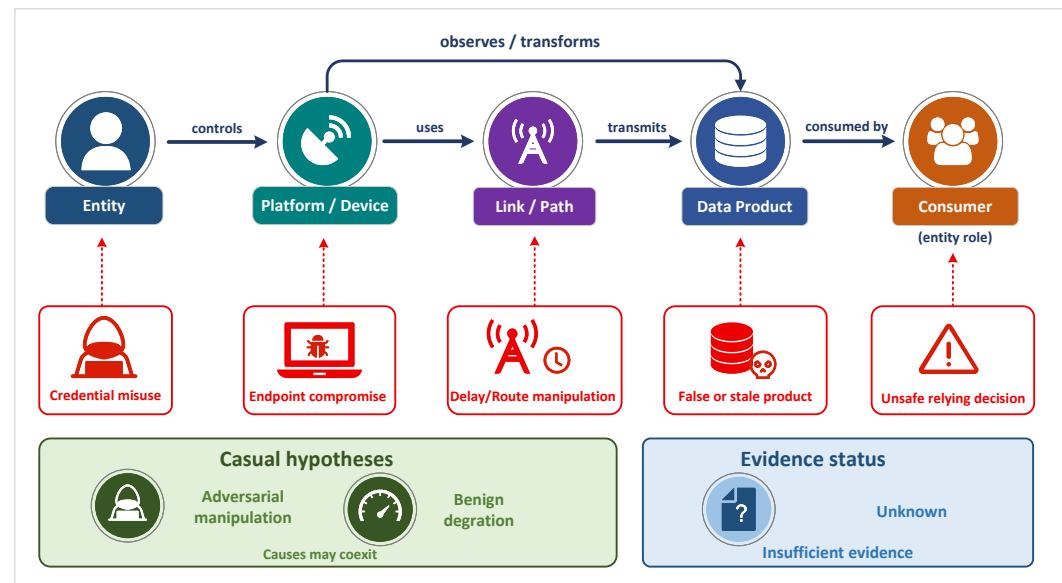


Figure 4. Simplified object dependencies and illustrative exposures. Solid arrows name operational relations; red dashed arrows associate an exposure with its object. The lower panels distinguish causal hypotheses from the epistemic status of the evidence. Causal propagation claims require explicit causal evidence and inference rules.

The graph supports evidence tracing and exposure analysis. A fresh attestation can support a scoped claim about measured platform state; verified provenance supports recorded derivation; corroborating observations support physical plausibility under their dependence assumptions. Each inference requires an explicit rule connecting evidence to a property. Hosting and transmission relations identify operational dependencies; inference about a property or causal effect requires a specified model. Compromised credentials may permit device access; a compromised device may alter observations; a manipulated path may delay delivery or substitute content if authentication fails; and a consumer may then act on the result. Figure 4 depicts this conditional chain. Traversal follows declared relation semantics to identify claims for reassessment; score updates depend on the specified inference model. If a causal or probabilistic model is used, its conditional structure, common causes, and parameters must be specified separately. Versioning and event ordering expose feedback cycles so that update rules can preserve the identity and dependence of recirculated evidence.

6.2. A Unified Threat-Analysis Sentence

Every scenario should be expressible as: *an attacker with capability C acts on object O, propagates through relations P, creates mission impact I, and is constrained or detected by controls K under assumptions A.* This grammar connects threat catalogs to assurance cases and exposes missing controls. Table 5 applies the grammar to common public scenarios at a capability level suitable for reproducible defensive analysis.

Three labels organize the interpretation of an anomaly. *Malicious* denotes evidence supporting intentional or adversarial manipulation; *benign-degraded* denotes a supported environmental or operational explanation; and *unknown* denotes insufficient evidence to resolve the explanation. Unknown describes the evidence status of an assessment. Operational states additionally include nominal operation and recovery, and malicious action can coincide with environmental degradation. For example, packet loss may justify restricting reliance on a path, while attribution of attacker intent requires separate support.

Table 5. Object-centered threat, degradation, and evidence-absence analysis.

Class and initial object	Capability or condition	Propagation path	Mission impact	Controls and residual uncertainty
Malicious/entity	Stolen or misused credential	controls device → endorses product → record accepted	Unauthorized tasking or false accountability	Short-lived delegation, multi-party approval, behavioral evidence, rapid revocation; valid credentials can still mask insider intent
Malicious/device	Capture, malware, or model replacement	hosts sensor/model → observes/transforms data	Consistent but false products across many consumers	Attestation, key protection, diversity, provenance, quarantine; measured components may omit sensors or runtime behavior
Malicious/link	Jamming, wormhole, selective forwarding, replay	transmits/custodies data → delays or substitutes version	Loss of availability, stale situational picture, biased fusion	Link context, authenticated sequence, path diversity, freshness policy; attribution remains uncertain
Malicious/data	False-data injection, poisoning, semantic manipulation	transformed / endorsed / recorded product → consumed decision	Incorrect classification, route, or resource allocation	Physical plausibility, independent modality, robust fusion, lineage; independence assumptions can fail
Benign environment	Fading, multipath, weather, occlusion, low energy	environmental conditions degrade link or observation evidence	Delay, disagreement, reduced coverage	Environment-aware baselines, graceful degradation, explicit “benign-degraded” hypothesis; models may be incomplete
Evidence absence	Disconnection, failed verifier, missing lineage, stale clock	prevents appraisal across several relations	Unsafe overconfidence or unnecessary isolation	Unknown state, bounded authority, conservative policy, later reconciliation; compromise attribution requires additional evidence

Recovery tracks review of cached data, keys, neighbors, and prior decisions after software is re-attested. Policies map the assessed properties, uncertainty, and recovery status to bounded actions—continue, reduce privilege, request corroboration, isolate, return, or escalate to a human.

6.3. The CTG Reasoning Loop

CTG-Trust is the context–temporal–graph workflow developed in this synthesis and shown in Figure 5. The workflow specifies inputs and decision records for implementations with selected and validated inference and policy mechanisms. First, evidence events are captured with source, subject, event and receipt times, clock uncertainty, context, value, and lineage. Second, evidence reliability and dependencies are appraised: a device self-report and an external verifier carry different evidential relationships. Third, a selected method assesses a named object property. Fourth, graph traversal identifies downstream claims requiring reassessment and shared causes beyond the local assessment. Fifth, the assessment is revised as needed and its uncertainty made explicit. Sixth, a mission policy chooses an action. Seventh, the decision, relevant versions, and evidence commitment are recorded for review. Eighth, independently appraised outcomes and later verification update the evidence base, with ground-truth labels established independently of the workflow’s decisions.

Evidence reuse requires special care. A gateway’s report can influence device reputation, link assessment, data quality, and a fused decision; treating the resulting scores as independent can count the same observation repeatedly. Implementations retain unique evidence identifiers, parent identifiers, and shared calibration, localization, model, operator, and relay dependencies. Exact duplicates contribute once to a given update. Distinct observations with shared causes require a joint model or a method with stated dependence assumptions; any discounting rule requires justification under those assumptions. For unresolved dependence, a policy may classify the evidence as potentially dependent or request another source. Statistical independence requires assessment of shared navigation references, training data, and environments, including across distinct identities or organizations. Ledger persistence preserves the recorded lineage for appraisal, subject to capture completeness.

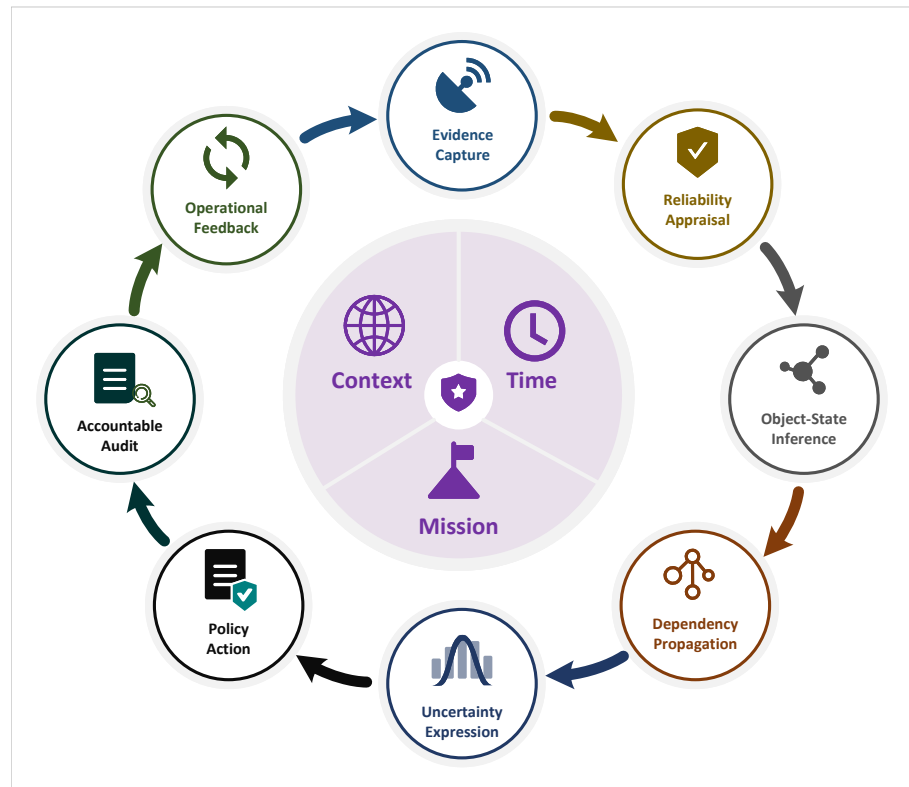


Figure 5. CTG workflow for evidence appraisal, dependency review, uncertainty representation, policy action, and audit. The diagram specifies information flow; performance depends on the selected methods and their mission-specific validation.

6.4. A Bounded Decision Trace

A constructed policy example illustrates how a USV gateway assesses a target-location product for UUV route replanning. Its thresholds specify the example; operational use requires mission-specific selection and validation. Policy version p_7 requires an unexpired delegation, verified product provenance, an accepted platform appraisal, acquisition age at most 60 s including clock uncertainty, and corroboration independent of the identified common localization failure. Replanning requires every gate to pass, with freshness and observation validity assessed alongside the ledger commitment.

At decision time, the gateway has a signed observation d_1 with an acquisition-age interval of $[42, 48]$ s, a fused product d_2 whose only observation parent is d_1 , and an observation d_3 from a second vehicle with age interval $[58, 66]$ s. The authorization, provenance, and platform-appraisal gates are assumed to pass. The lineage shows that d_1 and d_2 supply one observation lineage, while d_3 uses the same localization reference as d_1 . The decision trace is then reproducible: d_1 passes freshness; d_2 reuses the same observation lineage; and d_3 exceeds the worst-case freshness bound while retaining the shared localization concern. The gateway records the target-location claim as requiring further support and selects the preauthorized local fallback specified by p_7 in place of replanning. The decision rests on evidence sufficiency; attribution of malicious behavior is a separate assessment.

The audit record contains the gate results, lineage identifiers, assessed uncertainty, policy and platform-appraisal versions, delegated-authority basis, and selected fallback. A fresh observation with an adequately justified independent localization basis would trigger reassessment. Observation age and independence retain their acquisition and lineage meanings when ledger inclusion occurs later. This trace makes the framework operational at the level of a reviewable decision procedure while leaving threshold selection, fallback safety, and mission-loss evaluation to an implementation study.

6.5. Trust Methods and Appropriate Claims

The literature offers several method families, summarized in Table 6. Each expresses a different claim under different assumptions. The qualitative comparison provides a selection guide for future mission-specific validation.

Table 6. Trust-inference families and their role in CTG.

Family	Useful representation	Strength	Critical assumption/risk	Representative sources
Weighted multi-attribute score	Context-weighted evidence features	Interpretable and implementable at constrained edges	Weights and thresholds can be arbitrary or task-insensitive	[48,49]
Probabilistic trust/reputation	Evidence-updated belief or behavior propensity	Supports sequential assessment when the update model is specified	Priors, dependence, and behavior persistence require justification	[52]
Subjective logic	Belief, disbelief, uncertainty, and base rate	Makes ignorance visible and supports opinion combination	Dependence and source identity must be handled explicitly	[51]
State-space/Markov	Latent state with temporal transitions	Represents persistence and recovery	State meanings and transition parameters need mission data	[79]
Rule/tree/fuzzy methods	Human-readable rules or learned partitions	Can combine heterogeneous evidence	Coverage gaps and abrupt boundaries; calibration often unclear	[78,80]
Machine/deep learning	Learned anomaly or multi-view representation	Captures nonlinear patterns	Shift, poisoning, explainability, and overconfidence	[63,64,102]
Graph-based propagation	Relational evidence over dependencies	Connects local evidence to downstream exposure	Correlation, cycles, and collusion can amplify error	[53,81]

The underwater studies provide concrete examples of these model assumptions and their evaluation conditions. The HMM proposal in Arifeen et al. [79] maps packet-loss rate and packet-error rate to latent malicious/trustworthy states; its MATLAB illustration uses ten observations with randomly initialized probabilities and demonstrates a temporal construction under a synthetic numerical setup. TEUC combines data-, link-, and node-based evidence with a C4.5 tree and reward/penalty updates [78]. The study identifies acoustic loss as a reason normal nodes may receive low trust and assumes attack-free initial deployment so that the initial training evidence is unpolluted; its MATLAB simulation is conditioned on that setup. The broader UWSN review identifies evidence collection, evaluation, propagation, and updating as distinct design choices [80]. Together, these studies motivate three CTG requirements: keep node behavior distinct from sensor truth, assess attack classification alongside mission decision loss, and retain benign degradation as a possible explanation.

Equation (1) provides an assessment interface for these methods when the output states the assessed property, context, horizon, uncertainty representation, and supporting validation. Probabilities require an event definition and calibration assessment; ordinal scores require an ordering interpretation and justified thresholds. Spectral reputation, such as EigenTrust, expresses relative reputation; its use as a mission-success probability would require an explicit event model and calibration [53]. The reliance policy separately specifies authority, constraints, and decision consequences. Beta parameters, hidden-state transitions, scoring thresholds, message schemas, and performance targets remain empirical design choices for each implementation study.

7. Blockchain-Assisted Assurance: Roles and Boundaries

7.1. The Decision to Use a Ledger

The decision to use blockchain begins with the governance structure and required record semantics. A permissioned ledger is plausible when independently administered parties require joint control over accepted state transitions and governance constraints preclude assigning that role to one agreed custodian. The selection criterion is joint control of accepted state, assessed separately from participation count, signature use, and connectivity. A signed append-only log with independent witnesses, an authenticated database with protected audit records, or a jointly administered replicated database may provide the required accountability. The design should state which unilateral changes must be prevented, who can authorize conflicting updates, and which fault model the alternatives must tolerate before comparing their cost. Local verified control serves hard real-time loops.

A permissioned ledger is a plausible choice for the bounded-membership missions considered here because admission and endorsement can follow organizational authority. Permission model and consensus resource cost are separate design choices, with the consensus algorithm determining whether proof of work is used. Resource-intensive peers and ordering services can reside on capable shore, edge, or selected USV nodes, while constrained vehicles submit transactions or verify supported proofs under explicit trust assumptions. Hyperledger Fabric provides a representative architecture with identities, endorsement policies, channels, ordering, and validation [105,106]. Its documentation distinguishes crash-fault-tolerant Raft ordering from Byzantine-fault-tolerant ordering. Raft ordering provides crash-fault tolerance; tolerance of malicious orderers requires a consensus protocol with the corresponding Byzantine-fault assumptions. Ordering, endorsement, and application validation protect different steps; clients must check commit validity, since application-state updates require successful validation after block inclusion [106]. OAL accommodates another shared-state service when its fault, availability, and governance assumptions match. Consensus surveys describe trade-offs among throughput, finality, membership, energy, and fault tolerance [87,88,107].

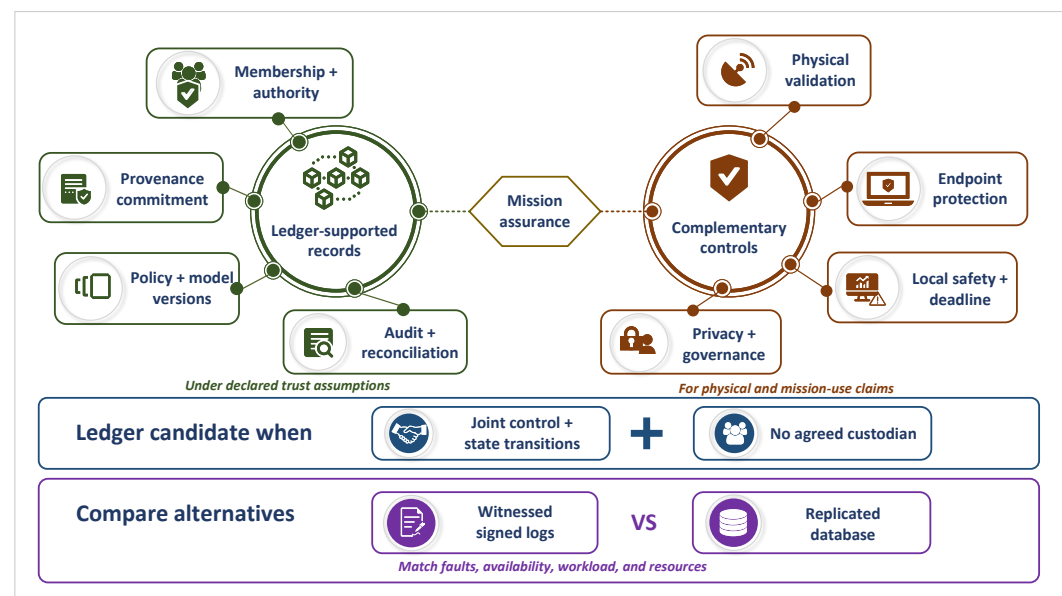


Figure 6. Blockchain boundary in mission trust. Ledger-supported records and complementary controls serve distinct claims. Selection depends on required joint control and comparison with witnessed logs and replicated databases under matched assumptions.

7.2. On-Chain and Off-Chain Partitioning

Raw video, sonar, point clouds, high-rate telemetry, detailed attestation evidence, and fine-grained location should remain off chain, encrypted and subject to data-use policy. The ledger records small, verifiable references: content identifiers or Merkle roots, producer and custodian identifiers, coarse timestamps or time ranges, schema and policy versions, attestation-result commitments, endorsement decisions, revocations, and audit outcomes. The commitment scheme specifies the exact bytes or canonical representation covered, its algorithm, and whether it commits to plaintext or ciphertext. Storage and retention arrangements preserve authorized access to the corresponding evidence; availability and recoverability depend on those arrangements. The consumer records the ledger state, local authority information, and time uncertainty available at creation and use. Reconciliation separately evaluates these against subsequently learned events.

On-chain hashes provide integrity commitments. Privacy comes from encryption, access control, selective disclosure, retention rules, key destruction, aggregation, and metadata minimization, because hashes can reveal equality, support dictionary attacks on predictable values, and bind sensitive metadata to a durable member-visible record. On-chain metadata also persists after an off-chain object is deleted. Privacy design therefore accounts for traffic analysis, membership visibility, timing, and the linkage of vehicle identifiers across missions. General IoT and blockchain reviews identify privacy and storage tension [35,36,108]; maritime governance adds commercial and jurisdictional concerns [32,33,109].

7.3. Five Design Patterns

Table 7 organizes recurring ledger uses by OAL claim, assumptions, and complementary assurance requirements. The patterns synthesize UAV, IoT, provenance, and attestation work into reusable design templates.

Table 7. Blockchain design patterns for cross-domain mission assurance.

Pattern	On-chain record	Assurance supported	Necessary assumptions	Complementary assurance required
Membership and authority registry	Organization, role, credential status, delegation and revocation version	Cross-party identity and policy-state consistency	Governed issuer set, protected keys, bounded offline credentials	Real intent, endpoint integrity, emergency authority conflict
Attestation anchor	Evidence digest, verifier identity, result, reference-value and policy version	Accountable device-state appraisal and freshness history	Trustworthy measurement root and verifier; adequate coverage	Sensor truth, unmeasured runtime behavior, compromised verifier
Provenance commitment	Data / version identifier, transformation and custody roots, endorsement	Tamper evidence for recorded lineage and processing	Adequate instrumentation, bounded time uncertainty, protected keys, retrievable evidence	Semantic correctness, omitted events, biased models, metadata confidentiality
Partition-aware audit and reconciliation	Signed sequence range, batch root, local policy/model version, reconnect event	Detectable gaps or conflicts relative to retained checkpoints	Protected append state, independently retained checkpoints, eventual evidence retrieval	Unwitnessed suffix loss, uncaptured events, uncertain event time, physical truth
Policy / model coordination	Approved artifact identifier, activation/rollback time, authorized endorsers	Agreement on approved versions among synchronized participants	Governed approval, version validation, activation and rollback rules	Actual deployment state, partitioned participants, artifact safety and performance

The membership pattern generalizes UAV identity and authentication proposals [27, 28,31]. It records which organization may issue each role, the applicable policy version, and how offline parties learn of revocation. Short-lived credentials, bounded delegation, and risk-specific offline envelopes constrain exposure while revocation information propagates.

UAV authentication prototypes provide evidence for membership and channel establishment. Tan et al. evaluate Hyperledger Fabric smart-contract operations and cryptographic communication/computation costs in a simulated 10 km × 10 km UAV mission

with cloud-hosted peers and assumed base-station coverage [28]. SwarmAuth combines PUF-based mutual authentication, a blockchain registry, and location-aware clustering, then evaluates security features, computation and communication cost, throughput, and delay using an EVM implementation and simulations [31]. OAL maps these results to entity/platform admission and ledger-supported registry claims. Observation validity, underwater partition behavior, cross-organizational governance, and consequence-weighted mission decisions form the corresponding cross-domain validation extensions.

The attestation pattern links RATS evidence appraisal to shared audit. An attester produces evidence; a verifier evaluates it under an appraisal policy; a relying party appraises the result for its intended use [41]. Recording a commitment to the result and policy supports later review and comparison of verifier outputs among organizations. Attestation architectures for blockchain and edge systems illustrate this intersection [42,43,110,111]. Sensitive measurement detail remains encrypted off chain by default. The permitted age of a result defines a relying-party policy bound. Continued reliance also requires assumptions about state changes after appraisal, a timing issue explicitly recognized by RATS [41].

Heterogeneous attestation is technically plausible, with platform-specific cost. The hardware-agnostic framework of Ott et al. [43] was implemented for TPM, AMD SEV-SNP, and ARM PSA evidence. Its mutual attested-TLS evaluation reported 31.2 ms for SEV-SNP virtual machines on the same host and 476.1 ms for the tested Intel firmware TPM, the latter dominated by quote generation. The latency measurements cover the SEV-SNP and TPM configurations; ARM PSA handshake evaluation requires an embedded TLS implementation. Cross-domain communication benchmarks additionally require representative links and mission conditions. These platform-dependent costs motivate retaining technology, configuration, freshness, and measurement scope with an attestation result. Measured software identity must also be interpreted against reference values and software assurance evidence; matching an approved image supports software-identity assessment, while vulnerability resistance and sensor accuracy require their respective assurance evidence.

The provenance pattern links W3C-compatible derivation to shared commitments. IoT provenance proposals use blockchain to protect provenance records or bind large off-chain collections [112–115]. For mission data, each derived product should identify immutable inputs, transformation code or model, parameters, responsible entity, and output version. Provenance supports explanation and accountability, including for AI artifacts [116]; observation-validity and mission-suitability appraisal assess the semantic quality of the transformation and its output.

The partition-aware pattern is especially relevant to underwater missions. A node or gateway keeps a signed, hash-linked local sequence, retains a batch root locally during disconnection, and submits it with a sequence range and prior checkpoint when connectivity permits. Verifiers check signatures, membership evidence, sequence continuity, duplicates, policy versions, and consistency with independently retained checkpoints. Inclusion proofs establish membership in a committed batch; capture completeness is a separate claim about the recording process. Missing entries inside a known sequence range or conflicting disclosed histories may be detected, while deletion of an unwitnessed suffix can remain invisible. Protected counters, independently held receipts, and explicit capture assumptions determine the strength of the audit claim.

Partition handling separately specifies ledger safety, ledger liveness, and local mission authority. Here, ledger safety means that correct replicas finalize mutually consistent histories under the selected protocol's assumptions; liveness means that eligible requests can progress when its quorum and communication conditions hold. Losing the required quorum can suspend new commits while previously verified records remain locally usable.

Under arbitrary partitions, the CAP result requires a trade-off between guarantees of linearizable shared read/write state and completion of every request at nonfailed nodes [117,118]. Local mission continuity and real-time deadlines depend on the applicable authorization and local execution conditions. Predelegated local actions can continue within their envelope, but an offline decision that requires fresh global state must wait, use an explicitly weaker state assumption, or follow its authorized fallback. Later reconciliation records conflicts and their disposition; the safety of an executed physical action depends on the conditions and controls at action time.

7.4. Governance, Revocation, and Recovery

Permissioned consensus places trust in explicit consortium rules. A consortium defines member eligibility, validator distribution, endorsement thresholds, emergency powers, software upgrade, key rotation, data jurisdiction, dispute resolution, incident disclosure, and exit. Replicas controlled by one administrator share administrative control of history. Assessment of organizational fault independence also accounts for shared administrators, cloud credentials, software defects, and key custody across validators. The governance assessment therefore identifies who can change membership and policies, override controls, or deny a quorum. Endorsement thresholds and validator placement must be evaluated against that distribution of authority and the required availability.

Revocation is temporal. A credential revoked with effect from 12:00 may have authorized a disconnected UUV's action at 11:55, while the gateway learns the revocation at 12:30. This interpretation requires a declared effective-time rule and trustworthy bounds on the action time. A signed timestamp binds a time assertion whose accuracy requires evidence about the clock and event capture. Audit separates event-time intervals, receipt and commitment times, locally known credential status, the applicable delegation, and subsequently learned revocation. If the event-time interval overlaps revocation or key compromise is suspected, the validity finding may remain disputed. Recovery reviews identities, affected models, data products, recorded provenance descendants, device re-attestation, and consequential mission decisions. The ledger and dependency graph help locate the recorded affected set; missing lineage requires additional investigation.

7.5. Real-Time and Safety Boundaries

Deadline-bound vehicle functions require local control with timing and safety arguments appropriate to stabilization, collision avoidance, or an emergency maneuver. The ledger can coordinate approved policy or software versions before operation, record summaries afterward, and support slower supervisory decisions, cross-party authorization, data release, and audit. Removing a ledger dependency from a control path avoids making consensus progress a prerequisite for that path, but platform safety still depends on local sensing, control, resources, and the chosen fallback. For example, emergency surfacing must itself be assessed against the local environment. Reduced audit freshness and cross-organizational coordination remain visible during ledger-service loss.

Figure 6 provides a claim discipline. Under the configured fault and key assumptions, a ledger supports evidence about accepted endorsements, versions, and record order. Physical, endpoint, timing, privacy, and governance evidence is additionally required to assess whether observations matched the world, endpoints met their integrity requirements, deadlines were satisfied, and authority was appropriate.

8. Evaluation Practices and Evidence Maturity

The literature includes conceptual architectures, analytical models, simulations, prototypes, datasets, and domain demonstrations. Each form answers a distinct question: secu-

rity analysis exposes protocol assumptions; network simulation explores scale; blockchain prototypes measure implementation behavior; and field trials examine integration under specified operating conditions. The evidence profile below organizes these forms as a basis for claim-specific evaluation of mission trust and future integrated OAL studies.

8.1. A Claim-Specific Evidence Profile

Figure 7 presents six complementary evidence categories spanning conceptual work, analysis, and operational experience. Category numbers identify evidence forms, while formal analysis, experimental control, environmental representativeness, and operational duration are assessed separately. Formal properties, field behavior, and resilience to a specified adversary each require direct support for the applicable claim. Each claim therefore receives an evidence profile identifying the methods used, system scope, assumptions, and conditions tested, with each category assessed within its own scope.

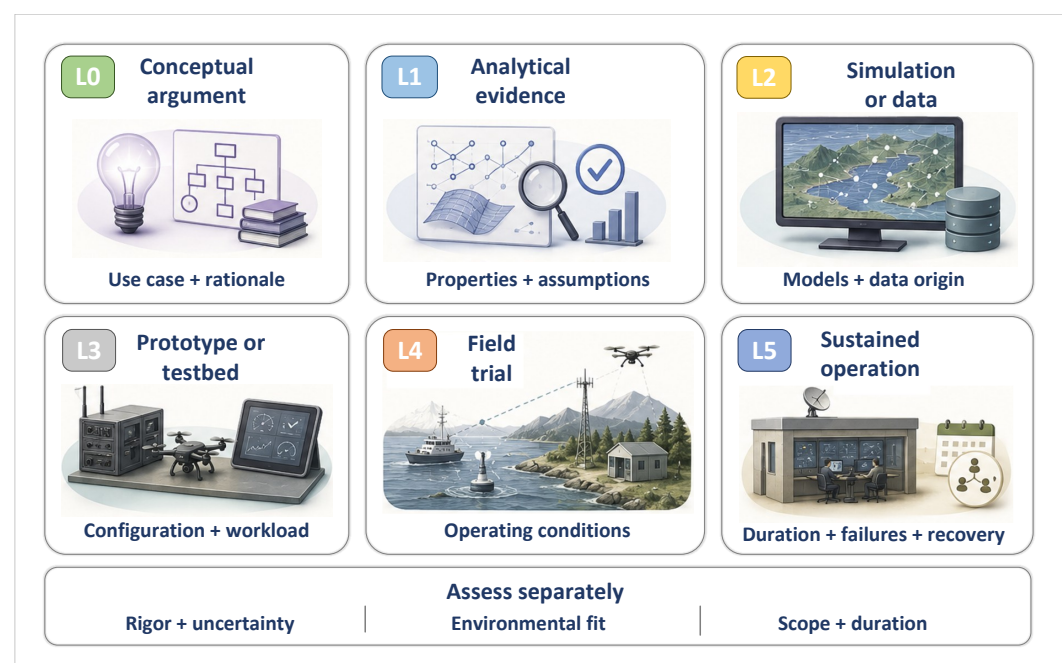


Figure 7. Claim-specific evidence profile. The six complementary categories indicate the setting and method supporting a claim. Verification scope and scientific rigor are assessed separately for each category.

Category L0 denotes a use case, taxonomy, or architecture supported by conceptual argument. L1 denotes mathematical, formal, or security analysis with explicit adversary and system assumptions. L2 denotes simulation or recorded-data analysis; the origin and representativeness of the data are recorded separately. L3 denotes an implementation tested in a prototype or testbed with a documented configuration and workload. L4 denotes trials under specified physical operating conditions, identifying which vehicle domains, communications, and participants were represented. L5 denotes sustained multi-party operation, with duration and observed governance changes, credential rotation, failures, audits, and recovery reported explicitly. A ledger-throughput test supplies prototype evidence for component performance; observation validity and mission safety require evidence directed to those claims. Integrated claims require evidence for each consequence-relevant dependency, including the interfaces between components.

8.2. Datasets, Simulators, and Testbeds

Existing public resources provide candidate components for a cross-domain evaluation environment. TON_IoT combines labeled IoT/IIoT telemetry, operating-system logs, and

network traffic collected in the UNSW Canberra cyber range; Edge-IIoTset uses a purpose-built seven-layer testbed and supplies baselines for centralized and federated learning [119,120]. Together with CICIOT2023 [121], they support device/link anomaly experiments. Transferring these experiments requires a separate model of maritime topology, physical degradation, and mission consequence. SeaDronesSee provides maritime UAV imagery and metadata for visual detection and tracking experiments [122]; evaluating observation validity additionally requires defined physical claims and appropriate ground truth. The supervised GNSS study of Semajski et al. [65] trains on synthetically manipulated signals and validates on two distinct real-world spoofing/meaconing datasets, contributing evidence for the evaluated signal features and recording contexts.

The original AirSim study describes a high-frequency physics engine, visual environments, and real-time hardware-in-the-loop support, with selected software components compared against real flights [123]. UUV Simulator extends Gazebo/ROS with hydrodynamics, thrusters, sensors, external disturbances, and multi-robot scenarios [124], while Aqua-Sim NG ports underwater-network functionality to ns-3 and evaluates the redesigned simulator architecture [125]. These resources address complementary components: air-domain perception and dynamics, underwater robot simulation, and underwater communication protocols. Coupling them is proposed integration work, requiring compatible software versions, time synchronization, shared geometry, validated channel interfaces, and reproducible event exchange. Attestation, provenance, consortium behavior, and mission-loss modules also require implementation and validation, with claim-specific ground truth established for each assessment.

Table 8. Evaluation resources and conditions for transfer. The OAL uses and metrics specify evaluation requirements developed in this synthesis; each resource supplies evidence within its documented scope.

Resource/practice	Primary evidence	OAL use	Important metrics	Required complement
CICIOT2023, TON_IoT, Edge-IIoTset	Labeled network, host, sensor, and attack telemetry; centralized/federated baselines	Device/link anomaly and uncertainty methods	Detection by class, calibration, false isolation, delay, missingness sensitivity	Cross-media maritime topology, physics, and mission-loss model
SeaDronesSee and GNSS records	Real maritime imagery/metadata and real-world signal-manipulation records	Observation validity, data-product quality, spoofing context	Detection error, geolocation error, uncertainty calibration, robustness to weather/viewpoint	Underwater, ledger, and additional observation contexts
AirSim	High-frequency physics, sensors, visual worlds, hardware-in-the-loop	Air-domain observation and mission use	Mission success, safety constraint violations, sensor/model shift	Acoustic-network and consortium-governance modules
UUV Simulator	ROS/Gazebo hydrodynamics, thrusters, sensors, disturbances, multi-robot scenarios	Underwater motion, sensing, control, integration	Localization, energy, mission completion, failure recovery	Field validation of hydrodynamics and sensor behavior
Aqua-Sim NG/ns-3	Underwater-network protocols, channel abstractions, routing experiments	Link/path evidence, delay, loss, routing and partition	Delivery, latency distribution, energy, path churn, outage duration	Physical observation-validity module
Permissioned-ledger testbed	Peers, ordering, endorsement, contracts, failures	Ledger/governance consistency and audit	Commit/finality latency, throughput, resource use, recovery, fork/conflict handling	Representative evidence sizes, event rates, and duty cycles
Cross-domain field exercise	Physical platforms, operators, weather, real links	Integrated lifecycle and mission-use suitability	Consequence-weighted mission outcomes, false quarantine, recovery time, audit completeness	Repeatable scenarios and systematic adversarial coverage

8.3. Metrics that Respect the Assurance Chain

An integrated mission-trust evaluation should cover five metric families. First, *inference quality*: discrimination where labels exist, calibration for probabilistic outputs or coverage for uncertainty intervals, uncertainty under missing evidence, detection and recovery

delay, and performance under distribution shift. Accuracy measures are complemented by the consequences of errors, including rare false isolation that can remove a critical relay. Second, *mission effect*: task completion, time, coverage, safety-constraint violations, decision regret, and resource allocation under adversarial and benign degradation. Third, *network cost*: bytes per evidence event, the full delay distribution, including its mean and tail behavior, energy, storage growth, duty cycle, and partition duration. Fourth, *ledger behavior*, where a ledger is used: endorsement and finality latency, throughput under realistic event batching, validator/resource cost, membership change, outage behavior, reconciliation completeness, and disputed histories. Fifth, *accountability*: percentage of decisions with resolvable provenance, time to determine affected descendants, policy/model version traceability, and successful revocation and recovery.

Experimental factors should cross object, threat, and context dimensions. Examples include compromised entities versus devices; isolated versus colluding nodes; jamming versus environmental fading; fresh versus stale attestation; independent versus correlated sensors; honest versus compromised gateways; short versus prolonged partitions; and routine versus high-consequence mission policies. A factorial or explicitly sampled scenario design can test their interactions. Ground truth should distinguish nominal operation, benign degradation, and adversarial intervention, including cases where they coexist. Missing or ambiguous evidence is a separate observation condition, and “unknown” is an inference outcome. Operational attribution requires evidence and validation appropriate to the deployment context, in addition to benchmark attack labels.

Training, threshold selection, calibration, and final evaluation should use separate data. Splits should respect shared recording episodes, platforms, sites, and time periods where these could leak information between partitions. Comparisons should use paired scenarios, with uncertainty intervals based on independent missions or simulation runs and with within-run packet dependence accounted for. Evaluation reports should state class prevalence, per-condition outcomes, the frequency and cost of abstention, and sensitivity to mission-loss weights. Latency summaries should include deadline misses and undelivered events so that delivery time and availability are evaluated together.

Baselines should also be layered. An authenticated centralized database and signed append-only logs test whether shared ledger state adds value for the stated governance problem; independently witnessed or replicated signed logs provide a stronger comparison where parties distrust one administrator. Removing replication, continuous appraisal, lineage, dependency handling, or explicit uncertainty in separate ablations can isolate their contributions. Comparisons should use the same application workload, evidence, access requirements, resource budget, and attacker capabilities. Architecture-specific mechanisms such as endorsement may differ; their trust assumptions and achieved guarantees must be stated. Each method should receive a comparable tuning budget and feasible operating configuration. Studies of Fabric-based access control and integrated IoT ledgers offer implementation starting points [112,126], completed by application-specific baselines.

Reproducibility requires the exact topology, channel and mobility models, workload, cryptographic settings, software releases and configuration, endorsement and fault model, random seeds, data partitions, trust-policy versions, and mission-loss function. Versioned provenance should cover the evaluation itself. If a policy threshold or model changes between runs, the reported outcome belongs to a different system configuration and assurance claim.

8.4. Interpreting Claims by Evidence Profile

UAV blockchain applications range from security and privacy architectures to swarm monitoring and resilient-network proposals [94,127–130]. Their evidence profiles depend

on the claim evaluated, the system configuration, and the reported test conditions. Broader work on Industry 4.0, federated learning, digital twins, supply chains, and government sharing supplies organizational patterns for comparison [131–135]. Their applicability to cross-domain missions remains conditional on endpoint, link, timing, governance, and mission-consequence evidence under intermittent connectivity.

Component results support distinct conclusions: authentication prototypes address admission and secure channels; anomaly datasets support classification under the represented attacks; and simulators characterize behavior under their configured physical and network models. Integrated mission assurance additionally requires validation of the interfaces between these components, including evidence freshness, shared failure causes, delegated authority, and decision consequences during disconnection.

9. Discussion and Research Agenda

9.1. Theoretical Implications

OAL represents trust as a structured relying relation. This yields three design implications. First, assurance arguments can be composed through typed dependencies when their inference rules and shared assumptions are explicit. Entity evidence supports authorization claims, device attestation supports claims about measured state, and observation and mission evidence support downstream use. Formal composition requires a theorem with explicit rules and assumptions for the combined claims. Second, trust is purpose-bound. Evidence sufficient for sharing low-sensitivity imagery may be insufficient for cooperative navigation, even when the underlying object is identical. Third, uncertainty requires an operational response: “unknown” records insufficient support for a conclusion and maps to a documented policy action. This epistemic status is represented separately from physical compromise and degradation.

The assurance chain also clarifies the relationship between provenance and truth. Provenance is an argument about history: who generated an artifact, through which activities, from which inputs. Observation validity is an argument about the world. Ledger consistency is an argument about shared records. Mission suitability is an argument about consequence. These claims require separate tests even when they share evidence or failure causes. Provenance and decision-accountability research supports traceable transformations [104,116]; OAL makes explicit the mission consumer’s judgment about which history supports which use.

9.2. System Resilience under Partition and Degradation

Future architectures should treat partition as a routine operating mode. Local policy envelopes require bounded capability, expiry, evidence-age limits, safe fallback, and explicit records of which global facts were unavailable. Reconnection combines synchronization with semantic reconciliation: the system determines which policy was locally available at event time, when a revocation became effective, which dependent claims require reassessment, and whether conflicting observations require mission review. Authorization under a previously delegated envelope and compliance with subsequently learned global state are distinct questions. Blockchain scalability reviews examine throughput and storage among other constraints [136,137]; cross-domain missions also require explicit treatment of long disconnection, sparse contacts, and consequence-sensitive local autonomy.

Resilience also requires functional diversity. If the USV is simultaneously acoustic gateway, time source, fusion node, policy-enforcement point, and validator, redundant vehicle count can conceal a shared failure dependency. The graph identifies vertices that support many mission paths; quantifying failure probability or mission loss additionally requires reliability and consequence models. Future work should compare architectural

redundancy (multiple instances) with evidential independence (different roots, media, sensors, and organizations). Two reports relayed through one gateway share transport and availability dependencies. Independently signed source observations may retain distinct origin evidence; whether the gateway can also alter or bias their interpretation depends on end-to-end protection, selection, and transformation roles.

9.3. Governance, Privacy, and Human Authority

Consortium governance is part of the security mechanism. Research should specify who can admit validators, approve reference values, activate contracts and models, invoke emergency powers, decide disputes, and audit the auditors. Maritime blockchain adoption studies already identify policy and organizational factors beyond technical performance [32,34,109]. In mission assurance, a governance failure can propagate like a cyberattack: an unsafe reference value makes compromised devices appear healthy, or a delayed revocation keeps stale authority active.

Privacy questions extend beyond encrypting payloads. Cross-mission identifiers, validator endorsements, timing, route custody, and model-use records can reveal organizational relationships or vehicle activity. Work on blockchain and IoT warns of immutable metadata and access-control tension [36,72,138]. Future designs should evaluate unlinkability, data minimization, selective disclosure, authorization-policy confidentiality, and deletion semantics alongside audit value. A mission may need accountable pseudonymity: an operator can verify that an authorized role acted while the long-term identity remains protected, and a governed process can resolve accountability after an incident.

Human authority must remain explicit. Automated trust inference can recommend reduced privilege, corroboration, or isolation, while high-consequence recovery and dispute decisions may require human review. The relevant evidence comprises model output, explanation, policy version, uncertainty, alternatives, and consequence. Authorized and auditable human override preserves this evidence trail and the accountability of the relying decision.

9.4. Testable Research Propositions

The following hypotheses translate the synthesis into comparisons for future studies. Each test requires a declared mission distribution, threat model, implementation, primary outcome, and minimum practically meaningful effect. Counterexamples and null effects delimit the conditions under which the proposed structure is useful.

P1 (assurance separation). In matched scenarios containing correctly signed but physically invalid observations, an implementation retaining separate observation, provenance, ledger, and mission-suitability assessments will have a lower unsafe-acceptance rate than a single-score implementation at a matched false-isolation cost. Both receive the same raw evidence, action set, and tuning budget. The outcome is the fraction of invalid products accepted for a specified consequential use, evaluated against an independent physical reference.

P2 (dependency awareness). For a specified binary event and prediction horizon, a probabilistic implementation that models shared sensor and recommender causes will have lower calibration error under correlated error and collusion than its conditional-independence ablation. Both use the same evidence and separately fitted parameters. Evaluation uses a prespecified estimator of held-out calibration, a proper scoring rule such as the Brier score, and decision loss, with results reported across correlation strengths and levels of lineage completeness. The test applies to outputs with an explicit probabilistic meaning.

P3 (diagnosis and uncertainty). Under mixed nominal operation, environmental degradation, adversarial intervention, and evidence outages, an implementation retaining competing causal hypotheses and an explicit unknown outcome will achieve lower pre-specified mission loss than a binary trusted/untrusted implementation. Both have access to the same evidence and operational actions. Separate ablations of causal diagnosis and the ability to abstain distinguish their contributions. The loss function includes corroboration delay, false isolation, unsafe acceptance, and lost availability, allowing the effects of diagnosis and response policy to be assessed separately.

P4 (bounded offline authority). Across prespecified partition durations and revocation schedules, bounded offline authorization will complete more permitted tasks before their deadlines than authorization that requires current global approval, and will incur fewer actions conflicting with effective global policy than unrestricted offline autonomy. Both outcomes and resource costs are reported as a trade-off curve over envelope duration and scope. Event-time local authorization and conflicts discovered after reconnection are evaluated separately, using explicit revocation-effective times and clock uncertainty.

P5 (selective ledger value). Under administrator equivocation or record withholding within a declared fault and network model, a permissioned ledger with independent validators will provide more complete, mutually verifiable audit histories than a single-administrator signed log. The comparison measures the fraction of previously accepted test events recovered, detection of inconsistent views, and time to establish an agreed history, using an independent event record as ground truth. A witnessed or replicated signed-log baseline tests whether any gain requires the ledger architecture. Communication cost and deadline misses are measured separately; substantive governance disputes require their own resolution procedures alongside record agreement.

P6 (evaluation coverage). A system configuration selected using transaction and network metrics alone will have higher held-out mission loss than one selected from the same candidates using observation, endpoint, governance, and mission outcomes in addition to those metrics. The comparison uses equal evaluation budgets, fixed selection rules, and paired held-out scenarios containing both benign and adversarial failures. The measured difference tests whether the added evaluation dimensions improve selection for that mission distribution.

Each comparison requires prespecified operating thresholds and analysis rules, effect sizes with uncertainty intervals, and reporting of null or adverse outcomes. Findings apply to the tested implementation and mission distribution; broader applicability requires evaluation across additional configurations and settings.

9.5. From OAL Cells to an Assurance Case

OAL can structure an assurance case when each high-consequence mission use is expressed as a scoped claim. For example, “fused product d_{17} is suitable for route replanning by consumer r until time t under weather class w and policy p .” The argument links that claim to required evidence across the layers. Observation subclaims identify the sensors, calibration, geometry, environmental model, residuals, and corroboration that support physical plausibility. Commitment subclaims identify protected input versions, producers, transformations, custody, and time. Where shared ledger state is required, ledger subclaims state membership, endorsement, ordering, and the agreed policy version. Mission subclaims connect freshness and uncertainty to an explicit consequence and permitted action. Missing support is recorded as an evidence gap; counter-evidence or a challenged assumption is recorded as a potential defeater. Both may justify restricting action, but their epistemic meanings differ.

This structure records both supported scope and boundary conditions. A verifier may establish that a measured software component matches an approved reference, with camera alignment and operator intent assigned to separate subclaims. A provenance record may establish the instrumented derivation path and identify a legacy transformation as a gap in completeness. A ledger may establish endorsement of a model version by three authorized peers, while safety and accuracy are supported by separate validation evidence. Pairing each guarantee with its conditions gives downstream consumers the exact scope of justified reliance.

Assurance cases must also be time-aware. Evidence has an observation time, issue time, receipt time, and expiry condition; a dependency can change between them. If a USV attestation is fresh when it receives a UUV batch but stale when it transforms and endorses that batch, the later activity requires an explicit continuity assumption or renewed evidence. Clock uncertainty is represented where GNSS is unavailable or suspected. For disconnected operation, the case records the last globally known policy and the local authorization envelope, preserving the scope of agreement available at event time.

Counter-evidence is first-class. A new sensor residual, revocation, incident report, or conflicting custody event should identify which claims and descendants become disputed. This enables proportional response: quarantine a derived track while preserving independently supported observations from a platform, or reduce a gateway's endorsement role while retaining its emergency relay function. Recovery closes the argument through review of affected credentials, software, models, data descendants, and mission decisions, producing a scoped and evidence-backed return to service.

Future tools could generate these arguments from the typed dependency graph while preserving human judgment. Policy authors decide evidence sufficiency, independence, consequence, and acceptable uncertainty. The generated artifact therefore distinguishes machine-verified bindings (signature, hash, sequence), verifier appraisals (measured state against reference), statistical assessments (calibration or anomaly), and governance judgments (waiver, dispute, emergency authority). A reviewer can then compare the conclusion with the strongest supported subclaim.

Finally, evaluation should measure the assurance case itself. Evidence coverage can be assessed over an independently reviewed set of consequence-relevant dependencies, with critical gaps reported individually alongside aggregate coverage. Seeded counterexamples test sensitivity to invalid arguments; formal soundness requires a proof under the stated inference rules. Operator studies can measure the time and error in identifying affected products; governance exercises can measure unresolved conflicts and time to an authorized decision. These measures complement detection and ledger performance by testing whether the structure helps users reach appropriately scoped relying decisions.

9.6. *Toward a Cross-Domain Unmanned Trust Profile*

The proposed Cross-Domain Unmanned Trust Profile (CDUTP) is an agenda for an interoperability profile: a small, versioned set of claims and evidence references for OAL boundaries shared by organizations. Its development agenda covers the schema, protocol bindings, and conformance tests. NIST Zero Trust Architecture supplies resource- and request-centered policy principles [39]; IETF RATS supplies attestation roles and semantics [41]; W3C PROV-DM and PROV-O supply entity-activity-agent provenance relations [44,45]. NIST's IoT device baseline informs endpoint capabilities [68]. 3GPP specifications for UAS support and maritime services provide communications and service context [139, 140]. Permissioned-ledger governance and endorsement provide candidate mechanisms for multi-party state coordination [105,106]. Conformance and interoperability require version-specific mappings and tests across the referenced architectures and specifications.

Software update evidence can reuse secure-update architecture. The IETF SUI architecture describes manifests, authorization, payload acquisition, installation, and device constraints [141]; CDUTP could reference an update identifier and installation result while keeping the firmware payload off chain. Applying similar bindings to learned models would additionally require model-specific compatibility and validation evidence. These mappings would connect domain claims to the mission assurance graph while preserving the scope of the underlying protocol guarantees.

A profile instance should minimally state: object identifier and type; claim and assurance layer; issuer and relying role; evidence or provenance reference; observed, issued, and expiry times; mission context and permitted use; uncertainty or appraisal outcome; applicable policy, model, schema, and contract versions; privacy label and disclosure conditions; parent dependencies; revocation reference; and on/off-chain binding. It should support disconnected issuance and later verification while labeling freshness at the relying time. Compact encodings are desirable for acoustic links, but semantic equivalence is more important than one wire format.

A staged standardization pathway begins with published use cases and claim-scope requirements, followed by a shared vocabulary and conformance tests for evidence lineage and version binding. Subsequent stages comprise plugfests across independently implemented UAV, USV, UUV, gateway, verifier, and ledger components; multi-organization privacy and governance evaluation, including member exit and disputed recovery; and alignment with relevant standards bodies. CDUTP provides a research agenda for that alignment.

9.7. Design Heuristics

Several design principles follow from the synthesis. Technology selection begins with the relying decision, object, and assurance claim, which define the roles of the trust method and shared-state service. Evidence management preserves evidence identity and common-cause information, stores raw evidence off chain, minimizes metadata, and explicitly represents staleness and unknown status. Architectures separate hard real-time safety from shared audit and treat policies, models, contracts, and reference values as versioned artifacts exposed to threats. Evaluation includes signed-log and replicated-database alternatives. Descriptions such as “proposed,” “analyzed,” “simulated,” “prototyped,” “field-tested,” and “deployed” identify distinct forms of support, with claim scope stated for each.

These heuristics guide technology selection. Signed append-only logs and replicated databases are candidate solutions when one accountable manager governs state and audit. Distributed agreement may be useful when independent parties require a common history, subject to the fault model, availability constraints, and comparison with independently witnessed logs. Evidence commitments require adequate key custody, time or sequence guarantees, and protected local state. Decisions with deadlines shorter than end-to-end communication require a local safety mechanism whose assumptions and fallback behavior are validated separately.

Responsible deployment adds one final boundary. Mission-trust infrastructure can improve safety, environmental observation, inspection, disaster response, and coalition accountability, while persistent identities and provenance traces also create surveillance and operational-security consequences. An assurance case should therefore include a legitimate-purpose claim, proportional data collection, retention and disclosure limits, and an accountable path for human appeal or incident review. Dual-use assessment focuses on capabilities and governance. Public reporting can disclose architecture, assurance semantics, failure modes, and evaluation methods while protecting credentials, exact operational configurations, exploitable topology details, and mission-specific tactics. This

balance supports scientific testing and responsible dissemination. It also reinforces the core OAL insight: governance artifacts and privacy policies are versioned, contestable evidence within the mission system and require the same accountable appraisal as technical artifacts.

10. Scope and Applicability of the Review and Framework

The review's conclusions concern the mechanisms and operating conditions represented in the cited literature. Differences in platform, threat model, workload, and evaluation outcome limit direct performance comparisons across studies. The synthesis supports architectural comparisons and conditional design recommendations; it does not establish the prevalence of particular methods or provide pooled estimates of their effectiveness.

UAV security, IoT reputation, road-vehicle trust, industrial provenance, and permissioned enterprise ledgers provide candidate mechanisms for cross-domain transfer. Their threat distributions, time scales, connectivity, energy, and governance define the conditions for comparison with air-surface-underwater operations. The synthesis identifies these transfer assumptions at the conceptual level. Target-environment validation is required to establish performance under the acoustic conditions and operational practices of a particular maritime deployment.

OAL, the dependency relation set, CTG loop, evidence profile, hypotheses, and CDUTP form a conceptual architecture for mission-trust analysis. Integrated implementation, formal composition analysis, and mission experiments provide the evaluation pathway for safety, calibration, availability, and audit-performance claims. Evaluation of the four object types should address coverage and consistent application across settings. Practical graph implementations require rules for abstraction, correlation, privacy, evidence expiry, cycles, and conflicting claims, together with measured computation and communication costs. The general assessment $T_o(t, c, m)$ accommodates several representations; a probabilistic implementation requires a defined event and horizon, an explicit statistical model, and calibration under relevant operating conditions.

The evidence categories organize complementary forms of support, with scope and rigor assessed for each claim. Tests of the research hypotheses require concrete comparators, loss functions, and scenario distributions to determine architectural trade-offs and effect sizes. Development of the CDUTP proposal entails a normative schema, interoperable implementations, and a conformance test suite. Formal analysis, controlled studies, field evidence, and multi-party governance exercises provide complementary support for assessing these contributions.

Finally, the public-evidence focus supports reproducible analysis of assurance, resilience, audit, and responsible governance in civil and publicly documented defense-support settings. Sensitive operational details remain within the governance and disclosure controls of the organizations that hold them. Future authorized studies can evaluate the framework under additional contested-environment and platform-specific conditions.

11. Conclusions

Air-surface-underwater cooperation turns trust into an end-to-end mission problem. An evidence-supported decision integrates a plausible observation, an intact and accountable provenance chain, a correctly governed shared state where one is needed, and evidence that remains suitable for the particular mission consequence. The OAL framework connects these four assurance layers to entity, platform/device, link/path, and data-product objects over the complete lifecycle from admission to recovery.

The proposed object-dependency graph and CTG loop organize evidence flow, shared failure causes, uncertainty, and policy action across that structure. They distinguish com-

peting explanations of system behavior from insufficient evidence and require numerical assessments to state their meaning and support. The evidence profile identifies complementary forms of analysis, implementation, environmental testing, and operational experience, with scope and rigor assessed for each claim.

Blockchain is a candidate coordination mechanism where independent organizations need shared membership and version state, attestation and provenance commitments, accountable audit, and post-partition reconciliation. Its value depends on governance, fault assumptions, and measured costs relative to signed logs and replicated databases. Physical corroboration supports observation validity; attestation appraises measured endpoint state; local safety mechanisms address communication outages and deadlines; and privacy and governance require additional controls. Mission suitability requires the evidence and conditions associated with each applicable component claim.

The proposed CDUTP and six research hypotheses define work toward interoperable, testable systems. The review contributes a structure for assurance questions and their connection to selected evidence, providing a basis for comparative performance evaluation. The next stage combines formal analysis, cross-domain testbeds, representative field trials, and multi-party governance exercises, measuring both decision consequences and the quality of the supporting evidence. The practical objective is to make each consequential mission use traceable to current, appropriately scoped assurance and explicit residual uncertainty.

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Abbreviations

AUV	Autonomous underwater vehicle
CDUTP	Cross-Domain Unmanned Trust Profile
CTG	Context-temporal-graph
DLT	Distributed ledger technology
OAL	Object-Assurance-Lifecycle
RATS	Remote ATtestation procedureS
UAS	Unmanned aerial system
UAV	Unmanned aerial vehicle
USV	Unmanned surface vehicle
UUV	Unmanned underwater vehicle
ZTA	Zero trust architecture

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